

THE KILLER SHOT

The deaths of four infants in Tamil Nadu within hours of being vaccinated raise concerns over the immunisation programme for measles and doubts over the efficacy of our healthcare system

■ By Amarnath K. Menon

Tamil Nadu enjoys the top slot among the Indian states in providing healthcare. It has a 95 per cent immunisation coverage. The World Health Organisation (WHO) has certified it as a tetanus free state for neonates. No polio case has been reported here in the last four years. Also, there has been no incidence of diphtheria and whooping cough in the state. Yet, it was here that a vaccine—which inherently holds the promise of continued good health—snuffed out the lives of four infants.

In a serious setback to the Government's vaccination programme for children, a measles vaccine—manufactured by Human Biologicals Institute (HBI), a division of the Indian Immunologicals Limited (IIL), Hyderabad—has raised fresh doubts about drug control norms, immunisation practices and paediatric care. The shocking deaths of the infants after they were given shots of the measles vaccine, sourced from Karchur primary health centre in Tiruvallur district of Tamil Nadu, have shaken the public confidence in community health services.

An embarrassed Chief Minister M. Karunanidhi immediately announced an ex-gratia payment of Rs 3 lakh each for the grieving families. In a swift response to the development in his home state, Union Health Minister Anbumani Ramadoss has directed an inquiry into the deaths and suspended the use of the vaccine manufactured by HBI while putting on hold the measles inoculation programme across the country. He has also directed health authorities to draw samples of the vaccine from primary health centres in Tamil Nadu for testing at the Central Research Institute (CRI), Kasauli.

While Orissa and Puducherry had banned the vaccine even before the Union Health Ministry got into the act, the Drug Control Authority (DCA) in



DEADLY VACCINE What made the vaccine toxic? Here are some probable reasons.

EXPIRED No. The vaccine was made in February 2008. It was due to expire in January 2010.

DEFECTIVE QUALITY Unlikely. Tested at CRI; certified by WHO. Vaccines with vial monitors have temperature-sensitive labels that change colour if the sample becomes unusable.

STORAGE PROBLEM Possible. Poor cold storage facilities due to frequent power cuts can spoil the samples.

DIRTY NEEDLES Possible. Deadly, if the syringes had been dipped in other chemicals (say, muscle relaxants) and were then reused to vaccinate the infants.

TAMPERED VIALS Reconstituted vaccine needs to be used within three hours. Bacterial growth occurs if kept out in the open. Also, used and new vials are kept in the same refrigerator. Callous handling may have caused the disaster.

SAFETY RULES If you have to take the vaccine, check if a fresh, disposable and sterilised syringe is being used. Also, look carefully for the expiry dates and ensure the vial is sealed.

"CONTAMINATION OF NEEDLES FOR VACCINATION COULD BE A CAUSE FOR CONCERN"

Andhra Pradesh raided HBI's manufacturing facility in Hyderabad and seized 11,750 vials of the vaccine. The company claims there is nothing wrong with its measles vaccine. What exactly did go wrong is unlikely to be known before mid May when the inquiry panel submits its findings. Their job will include testing or looking for vaccine potency, abnormal toxicity and sterility to check for efficacy and contamination besides the moisture content from random samples picked up from HBI. The Central Government, which procures and supplies the vaccine to the states,

has found that the questionable batch was made in February and is due to expire only in January 2010. But, its quality and safety could be suspect. Therefore, the Government is waiting for CRI's report.

However, if there is no lapse on the part of the manufacturer, the failing is somewhere along the delivery chain—preventable errors in storage, handling or administration. The vaccine may not have been stored properly at the right temperature if the cold chain was broken due to power cuts. State health officials contend that all primary health